

Introduction to Deep Learning

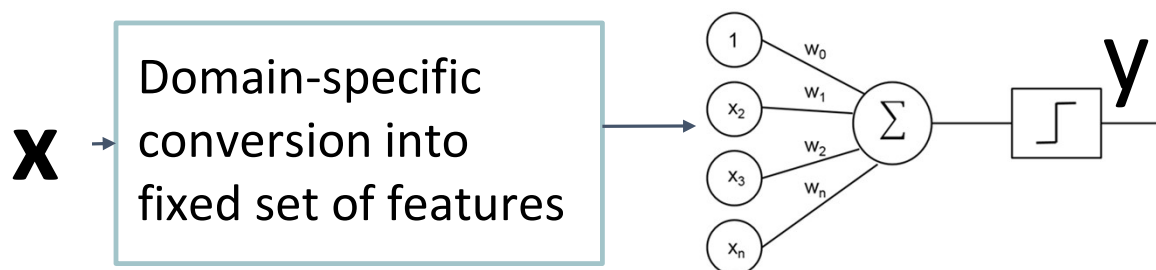
CS 217: Spring 2025-26

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Conventional Machine Learning



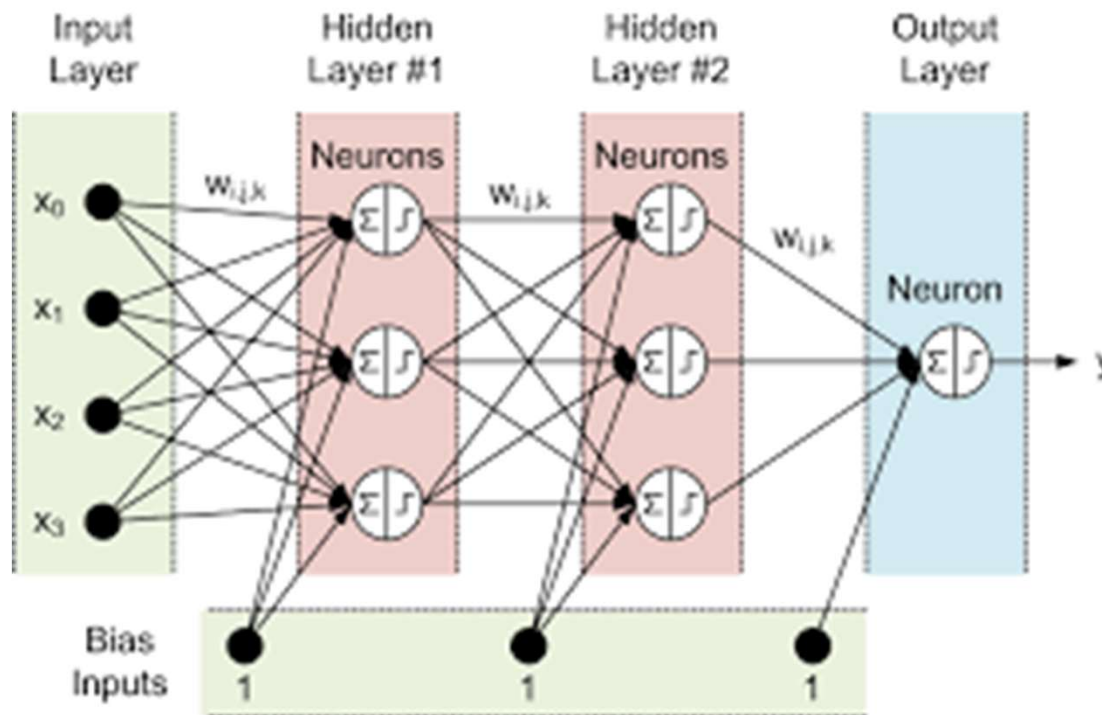
- Long history starting from the 1950s
- Many approaches: Perceptrons, Decision trees, Neural networks, Naive Bayes classifiers, SVMs



- ★ Convex training objective!
- ★ Reliable, scalable
- ★ Boosting, Bagging for more.
- ★ Many applications: search result ranking, retail, advertising, etc.
- ★ Mostly linear models, non-linear models based on kernels very expensive

Neural Networks

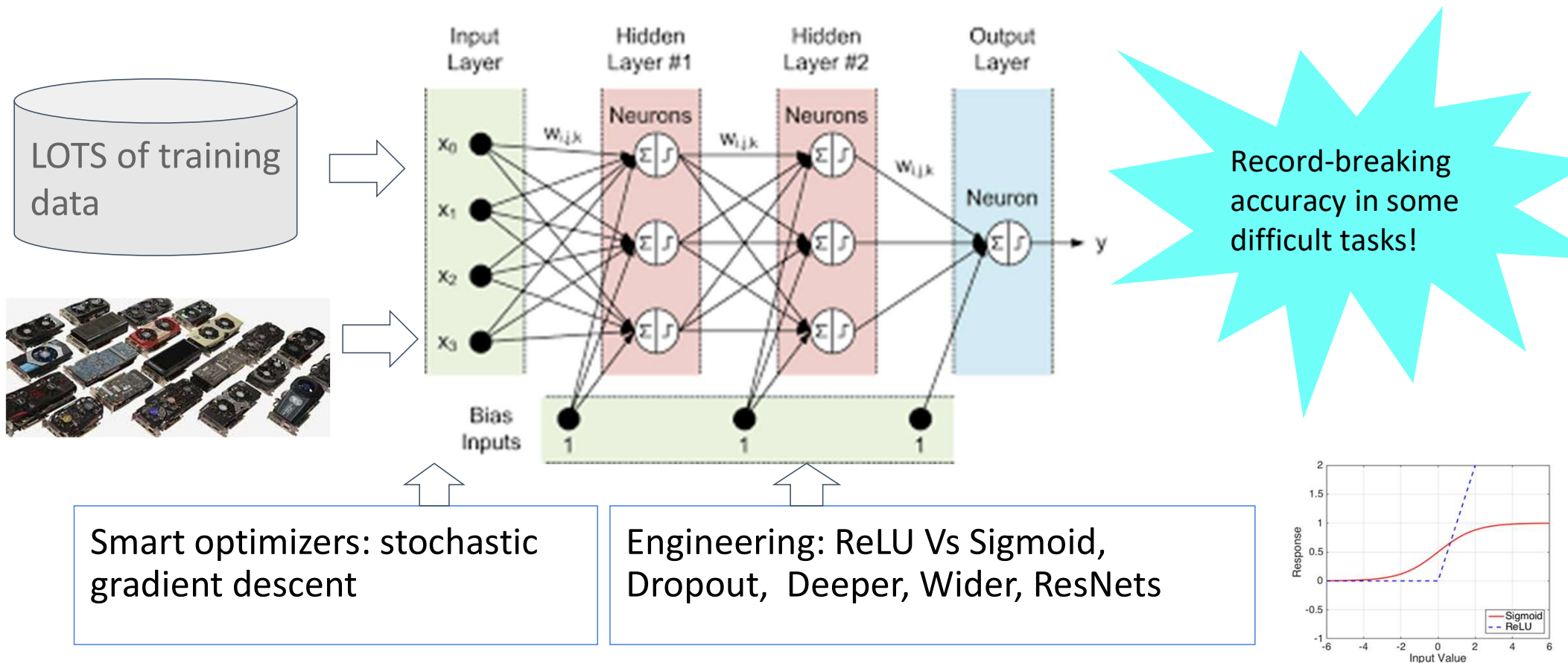
Many linear functions connected layer-wise after simple non-linear activation (Sigmoid)



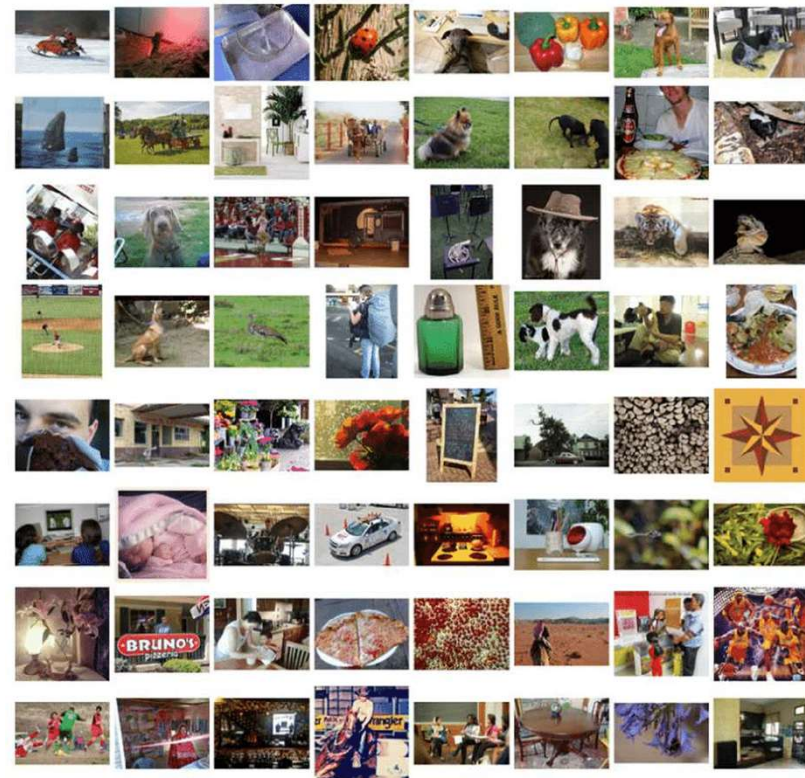
- ★ Universal function approximator
- ★ High-capacity
- ★ Automatically learns "multiple levels of representation, obtained by composing simple but non-linear modules that each transform the representation at one level into a representation at a higher, slightly more abstract level".

Neural networks considered too hard to train, too unstable, too slow.....until 2010

The deep learning boom (2011---)

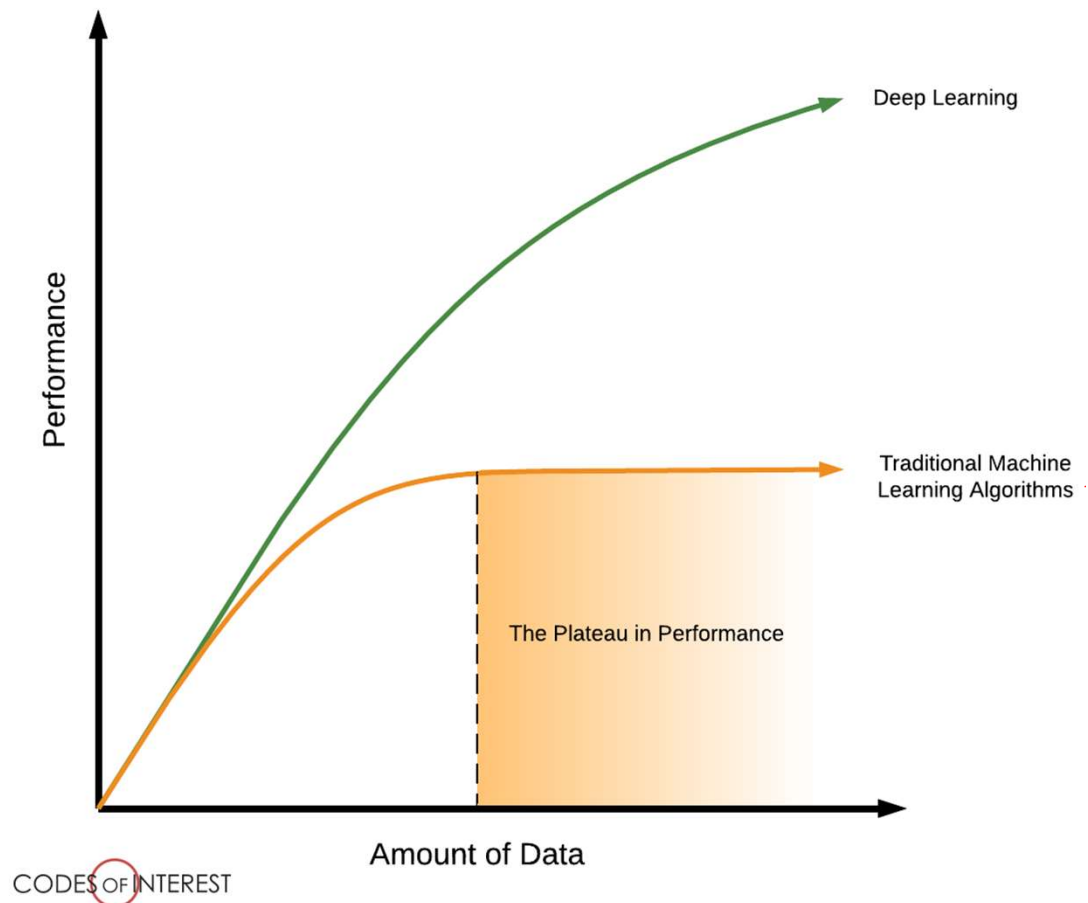


Success stories: vision



Traditional ML Vs Deep Learning

The ability to process huge amounts of data and continue to accrue accuracy gains from them.



Outline for this module

- Neural Architectures
 - Feed forward networks
 - Convolutional Neural Networks
 - Recurrent Neural Networks
 - Transformers
- Training a Neural network
 - Optimization algorithms, Hyperparameters
- Deep Learning in practice
 - Pretrained networks, fine-tuning, domain-adaptation